

VECTOR DATABASE + RAG TEST DOCUMENT

Topic: Smart Campus Energy Management

Purpose

This five-page sample document is designed to test a Retrieval-Augmented Generation (RAG) system that stores document chunks in a vector database and retrieves relevant passages to answer user questions. It contains policies, definitions, procedures, and numeric facts so you can test factual retrieval, multi-step reasoning, and grounded answers.

1. Overview

A smart campus energy management program helps a university reduce electricity waste while maintaining safety, comfort, and reliable learning environments. It combines occupancy sensors, smart meters, equipment schedules, and clear responsibilities for students, faculty, facilities staff, and administrators.

The program has four goals: reduce avoidable energy use; keep classrooms and laboratories comfortable; detect equipment that is running unnecessarily; and measure whether energy-saving actions are working. Energy reduction must never override safety requirements or equipment instructions.

2. Core Terms

Occupancy sensor: A device that estimates whether a room is in use. It may use motion, infrared, or other sensing methods. A sensor can fail to detect a stationary person, so it should not be the only safety control.

Smart meter: A meter that records electricity use at regular intervals and makes the readings available for analysis.

Automation rule: A condition and action, such as: if a classroom has been unoccupied for 15 minutes, turn off its lights, provided the room is not marked as an exception.

Exception room: A room where automatic shutoff is disabled or restricted because of safety, research, accessibility, or operational requirements.

Key principle: energy-saving automation must include manual override, exception handling, and a way to report faults.

3. Roles and Responsibilities

Students and staff should switch off lights, displays, and other permitted equipment when leaving a room. They should not unplug laboratory, medical, network, refrigeration, or safety-critical equipment unless authorized.

Faculty members identify teaching or research activities that require equipment to remain powered. They should report rooms where motion sensors may be unreliable, such as rooms where occupants sit still for long periods.

Facilities staff maintain sensors, smart meters, lighting controls, and automation schedules. They review fault alerts, investigate unusual consumption, and keep a record of repairs.

The campus energy coordinator reviews monthly consumption, approves policy changes, and publishes a short performance report. The coordinator must consult facilities staff before changing rules that affect laboratories or shared infrastructure.

4. Room Categories and Default Rules

Room type	Default automation rule	Important exception
Classroom	Lights off after 15 minutes of no detected occupancy	Do not shut off during an approved exam or event
Computer lab	Lights off after 20 minutes; computers follow lab schedule	Do not cut power to servers or active experiments
Corridor	Dim to 30% after 10 minutes of inactivity	Maintain legally required emergency lighting
Meeting room	Lights and display off after 10 minutes	Allow a booked meeting to extend the active period
Research lab	No automatic equipment power-off by default	Lab supervisor approval is required

These are sample policy rules for testing only, not real university regulations. A deployed system should use the institution's approved policy and local safety requirements.

5. System Architecture

A typical smart campus energy system contains sensors, a gateway or controller, a backend service, a database, a dashboard, and an alerting mechanism. Sensors provide observations; the controller evaluates rules; the backend records events; and the dashboard allows authorized users to inspect status and adjust permitted settings.

Suggested data flow

1. A room sensor reports motion or no motion.
2. The controller checks the room category, schedule, and exception list.
3. If the room has remained unoccupied for the configured interval, the controller issues an allowed action.
4. The system records the event with a timestamp and result.
5. Facilities staff can review failures or override the action.

6. Data and Privacy

Collect only data needed for energy management. Prefer room-level occupancy state over identifying individuals. Restrict dashboard access by role, protect device credentials, encrypt network traffic where supported, and define how long event logs are retained.

A motion sensor should not be treated as proof that a room is empty. False negatives can occur when people remain still, a sensor is blocked, or a device loses connectivity. For this reason, critical rooms need a safer policy, and automated actions should be reversible where practical.

7. Reliability and Failure Handling

If a sensor stops reporting, the controller should not assume that the room is empty. It should enter a defined safe state, raise an alert, and wait for a valid reading or an authorized manual action. If the network is unavailable, local controls should continue to work where the hardware supports it.

The system should record at least: room ID, event time, sensor status, rule evaluated, action requested, action result, and any override. Avoid storing personal identity unless there is a clear, approved need.

8. Example Event

At 14:00, Classroom C-204 reports no detected motion. The room's rule requires 15 continuous minutes without occupancy, so the controller waits. At 14:15, if the sensor is healthy, the room is not an exception, and no approved event is active, the controller turns off the ordinary lights and records the result. Emergency lighting is not affected.

9. Measuring Results

The energy coordinator should compare electricity use before and after a change, while accounting for room bookings, seasonal temperature, operating hours, and changes in student attendance. A simple before-and-after comparison can be misleading if the campus schedule changed at the same time.

Recommended metrics

Metric	Meaning	Example target for pilot
Energy use (kWh)	Electricity consumed in a period	Compare equivalent four-week periods
After-hours load	Use outside scheduled operating hours	Reduce by 10% without affecting safety
Automation success rate	Successful allowed actions divided by attempted allowed actions	At least 98%
False shutoff reports	Cases where a room was occupied when a shutoff occurred	Investigate every report
Sensor uptime	Time a sensor is reporting normally	At least 99% during the pilot

The targets above are illustrative pilot targets, not universal standards. Establish a baseline before judging performance.

10. Pilot Plan

Week 1 — Baseline: select two ordinary classrooms, verify meters and sensors, and record existing energy use and room schedules.

Week 2 — Configure: apply the approved classroom rule, test manual override, and confirm emergency systems are unaffected.

Week 3 — Observe: review event logs daily, interview room users, and fix sensor placement or schedule problems.

Week 4 — Evaluate: compare energy use with the baseline, review false shutoff reports, and decide whether to expand, revise, or stop the pilot.

11. Decision Rule

Expand the pilot only if energy use improves meaningfully, automation reliability meets the agreed target, no safety issue is identified, and room users can report problems easily. If energy savings conflict with safety or essential operations, safety and essential operations take priority.

12. Troubleshooting Guide

Problem	Likely checks	Recommended response
Lights stay on in an empty room	Sensor placement, timeout, sensor battery, room schedule	Check readings and configuration; do not shorten timeout without test
Lights turn off while someone is present	Blind spots, stationary occupants, sensor obstruction	Disable automation for the room until the issue is corrected
Dashboard shows stale data	Network, gateway, device heartbeat, backend status	Show the last-update time and alert facilities staff
Energy use increases after room schedule changes	Schedule changes, faulty relay, meter mismatch, sensor failure	Reinstall meter data and compare equivalent operating periods
A user reports an unsafe action	Room category, exception list, rule configuration	Stop the affected automation and escalate immediately

13. Glossary and Quick Reference

kWh: A unit of electrical energy. **False negative:** A sensor fails to detect a person who is present. **Manual override:** An authorized person temporarily takes control away from automation. **Safe state:** A predefined behavior used when reliable information is unavailable.

Sample policy facts to retrieve

- Classroom lights: 15 minutes without detected occupancy.
- Computer lab lights: 20 minutes; computer power follows the lab schedule.
- Corridor lighting: dim to 30% after 10 minutes, while preserving emergency lighting.
- Meeting room lights and display: off after 10 minutes, subject to booking rules.
- Research lab equipment: no automatic power-off by default; supervisor approval required.
- Sensor failure: do not assume the room is empty; alert staff and use the defined safe state.

End of document — use the questions below to evaluate retrieval accuracy, context grounding, numeric recall, multi-chunk reasoning, and refusal to invent unsupported facts.